

Assisting Colorblind Children: Nao's Role in Personal Style Development and Weather-Appropriate Dressing

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Abstract

This research paper presents the application of the Nao robot to assist colour-blind children in their daily lives. The objective is to empower these children by providing them with a supportive tool for making clothing choices and enhancing their style while considering weather conditions. The Nao robot, chosen for its friendly appearance and advanced camera capabilities, plays a pivotal role in this project. Using real-time image processing and

resource not only for this specific group but also for a wider demographic facing challenges related to colour vision. As ongoing project developments unfold, we anticipate the emergence of additional enhancements and valuable insights, which have the potential to create a significant impact on the lives of individuals living with this deficiency.

Index Terms: Human-Robot Interaction, Color-blind, Color Recognition, Clothing Recognition, Outfit-matching, Weather.

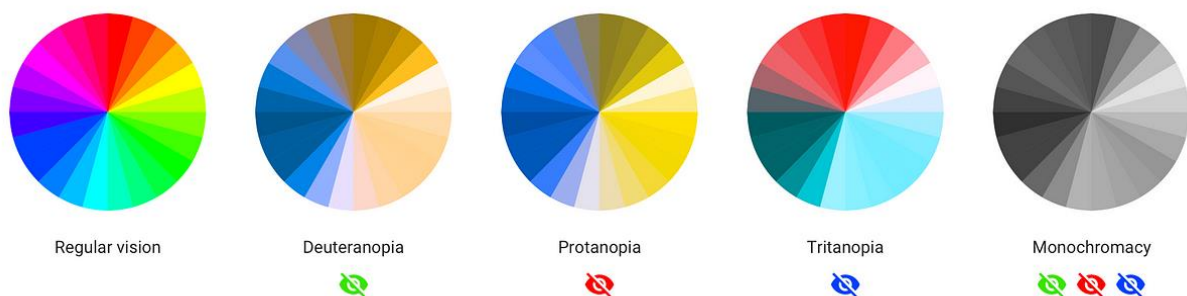


Figure 1 : Comparison of the perception of the color wheel by people with different types of color blindness

machine learning algorithms, the robot recognizes clothing colours and types. In addition, the system is integrated with a weather API, allowing Nao to match clothing choices with the weather and offer weather-appropriate clothing recommendations to the user.

The evaluation plan includes user testing, and data analysis to assess the system's effectiveness.

Our hypothesis suggests that the implemented system holds the promise of notably enhancing the self-assurance and self-reliance of colour-blind children regarding their dressing routines. Furthermore, its versatility positions it as a valuable

Introduction

Color vision deficiency is a condition in which a person sees colors differently than most people and has difficulties distinguishing between them. (1) Colorblindness presents unique challenges, extending beyond mere color perception and deeply impacting everyday activities, including clothing selection, especially for colorblind children. They manage this chore with the assistance of family members, by applying plastic braille labels or various sorts of stitching pattern tags to their garments, or by wearing clothes of a single color or without any patterns. (2) Medically speaking, there

is still no cure for color vision deficiency that runs in families, most of which adapt to live with it. (1) Our research is dedicated to harnessing the capabilities of robotics and human-robot interaction (HRI) to empower these children, facilitating independent clothing choices.

This paper introduces an innovative application of the Nao robot, specially tailored to support colorblind children in clothing-related tasks. Nao, known for its humanoid appearance and advanced camera functions, assumes the role of a friendly and interactive companion for these children. It utilizes real-time image processing and machine-learning algorithms to identify clothing colors and types swiftly and accurately. Additionally, the system seamlessly integrates with a weather API, enabling Nao to offer clothing recommendations aligned with the current weather conditions.

Central to our approach is voice recognition, simplifying natural and intuitive communication between the child and the robot. This creates a familiar and barrier-free interaction environment. Nao's dynamic movement capabilities add an engaging and interactive dimension to the experience, strengthening the bond between the child and the robot.

The Nao robot's advanced camera serves not only color detection but also object recognition, enabling the system to accurately identify various clothing types. This object recognition feature is pivotal in providing clothing suggestions that match the child's style.

Importantly, we have deliberately excluded the use of LEDs in our system design, recognizing the limitations of color perception. This careful feature selection ensures a seamless and effective interaction, ultimately empowering colorblind children to make informed clothing choices, fostering their independence and self-assurance.

In the complex intersection of robotics, HRI, and assistive technologies, our project represents a significant step towards promoting autonomy and enhancing the self-esteem of colorblind children. We are confident that this research has the potential to improve the quality of life for colorblind individuals and inspire future developments in assistive technologies for those with visual impairments. This paper delves into the design, implementation, evaluation plan, and the potential

impact of our Nao-based system, emphasizing its importance in the field of human-robot interaction.

Related works

The utilization of robot-based systems to assist specific populations has seen a notable increase. Social robots are gaining traction due to their capacity to address various social and psychological challenges. Engineered to engage with humans in a social and empathetic manner, these robots find application in diverse areas such as healthcare, education, and mental well-being. This study focuses on the potential of social robots to aid children with color vision deficiencies.

In a previous study by Yang et al., (2) a camera-based system was presented, aiming to enhance the quality of life for visually impaired individuals by enabling the differentiation of garment patterns and colors. This system, departing from conventional texture analysis methods, introduced the concept of the "Radon Signature" to categorize clothing patterns into four groups (plaid, striped, pattern less, and irregular) and recognize 11 distinct colors.

Voysey, I. et al. (3) concentrated on visually impaired children from a human-interaction perspective, shedding light on the integration of technology, such as social robots, into the lives of children facing unique challenges. Their research delved into the initial interactions between children and robots, investigating how these early experiences shape children's perceptions and engagement with this technology. Moreover, their work underscored the significance of tactile exploration, speech, and interaction modalities for children with visual impairments [3].

Drawing from Ros et al.'s propositions (4), the study highlighted two major challenges in child-robot interactions. Firstly, the issue of age and social significance was addressed. Young children readily attribute social significance to artificial agents and interact with them accordingly, likely due to the prevalence of imaginative play as a tool for learning and exploration during early childhood. This trend, however, is less evident among older children. Secondly, concerning the system's interactive complexity, it was found that engaging young children could be effectively achieved with basic and responsive behaviors such as giggles. However, this approach did not yield the same results with older children. Consequently, it is reasonable to assume that to captivate older individuals, a robot would

need to exhibit more complex behaviors and a broader range of social responses.

Upon conducting a comparative analysis of these related works, it is apparent that both studies converge on the central theme of human-robot interaction, but they diverge in their interpretations of certain aspects. This underscores a common observation in the field of Human-Robot Interaction (HRI): user expectations tend to escalate in direct proportion to the sophistication of the robot, thereby raising the possibility of user disengagement upon encountering the system's limitations. (4)

This initiative, being novel, marks a step toward increasing autonomy and enhancing the self-esteem of color-blind children in the complicated convergence of robotics, HRI, and assistive technology. We are convinced that our research will improve the quality of life of color-blind children and inspire future innovations in assistive technologies for those with other forms of visual impairment. This study discusses the design, implementation, and evaluation plan of a Nao-based system.

Design

Our design approach is informed by the distinctive needs of color-blind children, with a keen focus on leveraging Nao's capabilities effectively. Our choice of Nao – a 58 cm humanoid that features two legs, two arms, two hands, and a head with expressive LED-colored eyes (5)- was well thought out. Unarguably, its babyish appearance (also referred to as baby schema) establishes a sense of familiarity and approachability, vital for interactions with children.

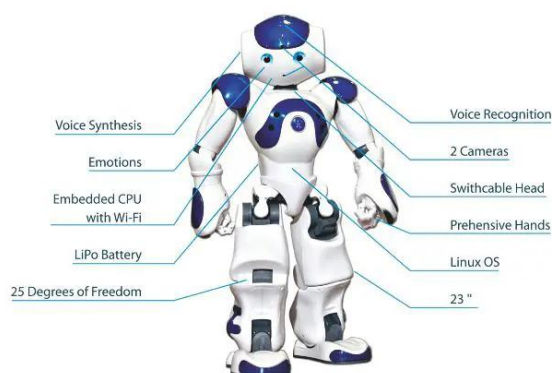


Figure 2: Nao's features

The core of our design centers on the two 5-megapixel Omni Vision cameras of the Nao robot, serving the dual purpose of color detection and object recognition. This versatility enables us to not only swiftly and accurately identify colors but also distinguish different clothing types, aligning perfectly with our goal of assisting colorblind children in their clothing choices.

Integration of a weather API is a pivotal component of our design, adding a practical layer to our system. Nao not only suggests clothing options but also considers the current weather conditions via the internet. This feature ensures that clothing recommendations are not just visually appealing but also suitable for the prevailing weather conditions.

Our decision to prioritize voice recognition as the primary means of communication is rooted in the aim of creating a natural and intuitive interaction. We understand the preference of colourblind children for familiar, unencumbered conversations, and voice recognition provides a direct and seamless channel for communication.

Notably, Nao's movement and LED sensors were intentionally ignored in the system design, acknowledging the limitations of color perception by the target user in interpreting color from the LEDs. Again, although capable of movement, we did not utilize the motion sensors as that function was not required for the scope of the interaction. This deliberate feature choice enables smooth and effective interaction, ultimately achieving the goal of the research. Overall, the design allows the robot to interact with the user and reliably distinguish clothing items, match the choice with the weather, and give clothing recommendations that are appropriate for the child's style.

In essence, our design demonstrates a profound understanding of the specific requirements of colorblind children. Through the Nao robot's humanoid appearance, advanced camera capabilities, weather awareness, voice recognition, and the exclusion of LEDs, we aspire to create an innovative system that not only helps but also fosters independence and self-assurance in clothing selection. This meticulous design ensures practicality, effectiveness, and user-friendliness, tailored to the needs and preferences of our target audience.

Implementation

In this project, we are primarily interested in color recognition. Before letting the Nao recognize a particular cloth type, we want to select those with the specified color. To do so, we are using OpenCV, which is a software library used for real-time computer vision. We are using the python version since we are using python to program Nao in Choregraphe.

To detect a certain color in a source image, we can use the function “inRange” from OpenCV which, given an upper and a lower limit in the form of an RGB triplet, can generate a mask to select the color between the two limits:

```
Llimit = np.array([98,50,50]) # lower limit array
Ulimit = np.array([139,255,255]) # upper limit array
# generate color mask
colorMask = cv2.inRange(source, Llimit, Ulimit)
```

The last step is to select only the colors indicated by the mask in the source image with the function “bitwise_and”:

```
colorImage = cv2.bitwise_and(source , source,
mask=colorMask)
```

In the example below, we wanted to detect the blue color of a sock. To do so, we set the lower limit to (5, 20, 110) and the upper limit to (105, 166, 205). These limits have been manually tested and modified to select as accurately as possible the color of the sock.



Figure 3: The unfiltered image of our blue sock

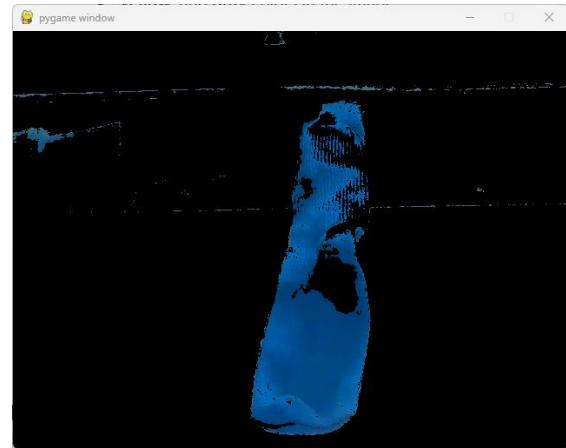


Figure 4: The filtered image where the mask cuts away all the colors except the blue ones

The result is not perfect since we are filtering the image only once. We believe that we could have better results with two consecutive filtering passes.

Unfortunately, we were not able to test this algorithm on a Choregraphe simulation nor on the real Nao because the version of python on them, the version 2.7, is deprecated and the modules supporting this version are hard to find and do not work properly anymore. Moreover, we encountered problems where the Nao had modules that Choregraphe did not and vice-versa. We were only able to test the code using a standard installation of python 3.11 with pygame to control a laptop webcam to take pictures.

The research by T. L. Nguyen et al. [5] stands as a testament to the successful integration of human expertise and perception into the robotic color recognition process, effectively leading to a robust system with the ability to accurately identify colored objects. Leveraging a Sugeno-type fuzzy system, the researchers skillfully integrated human-defined linguistic labels and expert knowledge into their color recognition framework. What is more, their use of the Evidence theory for information fusion from multiple cameras significantly bolstered the system's overall reliability and accuracy, providing a nuanced understanding of perceived colors.

Comparatively, our project focuses on employing OpenCV for color detection via masks based on defined color ranges. While we faced challenges testing it on the NAO robot, our testing on a standard laptop webcam highlighted that the accuracy of color detection hinges on factors such as lighting conditions and the quality of the webcam. Unlike the research paper's method, which uses

specific cameras for detection, our approach offers a more versatile and generic solution applicable to various platforms.

With this in mind, our project's expectations hinge on the accuracy of color detection in images captured by the webcam, which may fluctuate based on real-world conditions. However, it's important to note that the information fusion technique employed in the research paper of T. L. Nguyen et al. [5] demonstrated significant enhancements in color detection rates. Therefore, if we choose to implement a similar information fusion method with OpenCV, it has the potential to improve the precision and reliability of your color recognition system. The success of this integration will depend on factors such as the quality of image sources, the chosen fusion method, and how uncertainties are managed. This comparative analysis serves as a valuable guide for understanding the distinctions between the two approaches and setting informed expectations for our project.

Continuing with the script representing a comprehensive solution designed to establish a connection with the Nao robot, leveraging its vision module to capture images for clothing detection using a YOLOv5 model.

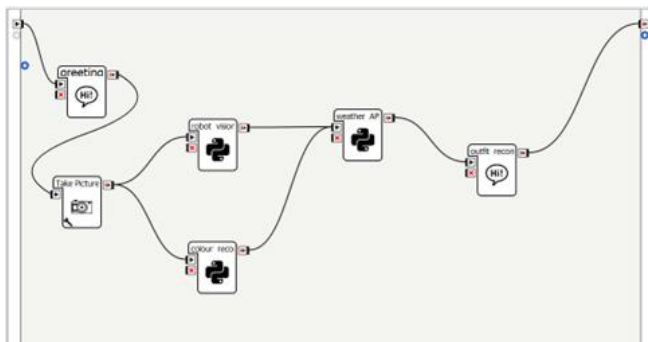


Figure 5 : Figure of Choregraphe

Firstly, the script initiates a connection to the Nao robot using the qi library, enabling smooth communication between the script and the robot. Subsequently, it activates the robot's vision module, subscribing to the video stream for continuous image acquisition. Notably, a pre-trained YOLOv5 model is loaded to facilitate efficient and accurate clothing detection within the captured images.

The script's central component revolves around the implementation of the `perform_clothing_detection` function. This pivotal function processes the

acquired images, subsequently leveraging the YOLOv5 model to identify and detect various clothing items within the images.

```
def perform_clothing_detection(Image):
    # Load the image
    img0 = Image.open(Image).convert('RGB')
    img = torch.from_numpy(img0).to(device)
    # Inference
    img = img[None] # Add a batch dimension
    pred = model(img)[0] # Model inference
    # Apply non-maximum suppression to get the
    most confident predictions
    pred = non_max_suppression(pred,
    conf_thres=0.25, iou_thres=0.45)
    # Display the detected clothing items
    for det in pred[0]:
        if det is not None and len(det):
            for c in det[:, -1].unique():
                n = (det[:, -1] == c).sum()
                print(f'Clothing class {int(c)}: {n}
    detections')
```

A continuous loop is then initiated to ensure a consistent image capture process from the vision module. The captured images are saved for further analysis and storage. Concurrently, the script employs the YOLOv5 model to identify clothing items and takes appropriate actions based on the detected objects, thereby showcasing the practical application of the implemented script in real-time scenarios.

Furthermore, the script is designed to handle unforeseen interruptions gracefully. A built-in mechanism for `KeyboardInterrupt` is incorporated, ensuring a smooth and controlled exit from the script. Before terminating the execution, the script unsubscribes from the video stream and performs necessary cleanup operations to maintain the system's integrity.

It is essential to note that the successful execution of the script relies heavily on the installation of pertinent Python packages, including `qi`, `torch`, `PIL`, and the YOLOv5 dependencies. The YOLOv5 object recognition library, developed by Glenn Jocher, has proven to be an invaluable asset, providing the necessary framework and tools for efficient and accurate clothing detection within the implemented script. The script's dependencies and the utilization of the YOLOv5 library are vital components that contribute to the script's overall functionality and effectiveness.

It's important to note that the script, as of the current state, is not fully functional and should be considered a work in progress. While the code and its components have been meticulously designed and documented, ongoing modifications and adjustments are required to ensure its operational readiness. This script serves as a foundational framework with the potential to deliver robust clothing detection capabilities on the Nao robot, but further refinements and testing are essential to achieve its intended functionality. As development progresses, we anticipate resolving any existing issues and optimizing the script to fulfil its purpose effectively.

In parallel with Nao's ability to recognize and categorize clothing items using OpenCV and object recognition, the system is further enriched by its capacity to provide personalized clothing recommendations based on real-time weather data, thus fostering a more dynamic and interactive user experience.

The objective of this part is to establish a connection with an external weather API, specifically the OpenWeatherMap, to acquire real-time weather data. This capability enables Nao to offer tailored clothing recommendations to users based on the prevailing weather conditions. The code implementation consists of two key components.

In the first part, it configures an API call to OpenWeatherMap using the Python "requests" library. The "requests" library facilitates the HTTP request process to the API endpoint. Subsequently, the code awaits a response from the API. The received HTTP status code is examined, with a status code of 200 denoting a successful connection and response. Other status codes signify issues related to the call itself or server-side errors, such as the HTTP error codes 400, 401, 402, or 403.

```
import requests
api_key = '789157358b584ea3b7ede5c37897105d'
city = 'Edinburgh'
url =
'http://api.openweathermap.org/data/2.5/weather
?q='+city+'&appid='+api_key
response = requests.get(url)
if response.status_code == 200:
    data = response.json()
```

Upon receipt of a valid response, the code proceeds to extract pertinent data from the returned JSON payload. These data components encompass the

current temperature, weather description, and the "feels-like" temperature. The temperature, initially presented in Kelvin, is meticulously converted into the Celsius scale. Furthermore, the weather description is categorized into one of three predefined categories. Category 1 corresponds to rainy or windy conditions, Category 2 is assigned to snowy weather, and Category 3 represents favorable weather conditions.

```
temp = data['main']['temp']
temp = temp - 273.15
desc = data['weather'][0]['description']
feeling = data['main']['feels_like']
feeling -= 273.15
if desc.find("rain") == 1:
    category = 1
elif desc.find("snow") == 1:
    category = 2
else:
    category = 3
print('Temperature:',temp,' C°')
print('Description: ',desc)
print('Feeling like: ',feeling,' C°')
```

The selection of the OpenWeatherMap API as the data source for this project stems from several considerations. Firstly, the API provides global weather coverage, ensuring that users worldwide can benefit from the weather-advisory feature. Additionally, the API offers a free plan, replete with generous usage limits that comfortably accommodate the project's requirements. The decision was further facilitated by the team's pre-existing familiarity with the JSON structure of the data received from the OpenWeatherMap API, streamlining the integration process.

As demonstrated by the code snippet, it connects to the OpenWeatherMap API using an API key and fetches weather information pertinent to the city of Edinburgh. This code serves as a practical example of how the project can collect and process real-time weather data. Furthermore, the integration of this capability into Nao's functionality provides a contextually rich and user-specific clothing advisory system, thereby augmenting Nao's capacity to engage users effectively. This implementation is envisioned to have broader implications, particularly in enhancing user interaction by offering personalized and timely guidance based on prevailing weather conditions.

Research protocol

The development of our project followed a systematic and evolving process, with each step carefully considered to address the specific needs of colorblind children in clothing selection. It all began with the selection of the project itself, aimed at aiding colorblind children in their clothing choices. The goal was to create a project with an impact on human life. Next, the choice of the robot played a pivotal role, with the Nao robot being chosen for its humanoid appearance and advanced capabilities in detection and communication, facilitating friendly interaction with children.

To inform our approach, we embarked on a research phase by consulting relevant academic papers. These studies guided us in determining the best methods, including the use of computer vision (OpenCV), for color and object detection. OpenCV, an open-source library, became the cornerstone of our technological approach, enabling the Nao robot to identify the colors of children's clothing.

The evolution of the project included a crucial addition - the consideration of weather conditions in the robot's clothing recommendations, thereby establishing a direct link between human-robot interaction and children's daily needs. For this, we explored new methods and delved into academic papers to incorporate features such as real-time weather data retrieval through an Internet connection and object detection to understand what the child presents to Nao, so appropriate recommendations can be made based on weather forecasts.

The testing phase proved to be critical. Initial trials with the Nao robot allowed our team to become familiar with the robot and adapt to its features. Subsequently, coding was undertaken to implement the functionalities we had conceptualized. The final phase involved further testing, where we could assess the effectiveness and limitations of our code and produce a demonstrative video of the system in action.

Evaluation Plan

In this research, the core question is: **"What is the impact of our Nao-based system on promoting social interaction among color-blind children while aiding them in choosing clothing that aligns with their style and is weather-appropriate?"**

Within the constraints of practicality and time limitations, we've tailored an evaluation plan to comprehensively assess the functionality of our Nao-based system. Our evaluation process kicks off with internal testing sessions, aimed at simulating real-world scenarios involving colorblind children and the Nao-based system.

These sessions entail the execution of specific scenarios to gauge the system's responsiveness to user commands and its adaptability across varying conditions. We actively collect qualitative data during these sessions, observing user interactions, facial expressions, and verbal feedback, with a focus on user engagement and satisfaction.

Key performance metrics, including clothing color detection accuracy and the relevance of weather-based recommendations, will be measured to evaluate the system's functionality. We'll also examine the frequency and nature of user interactions to gain insights into engagement and system effectiveness.

Moreover, our evaluation plan incorporates the use of surveys to gather comprehensive feedback from colorblind children. Drawing from insights in existing research, we aim to create engaging and age-appropriate surveys to understand the system's usability and impact from the user's perspective.

The approach to survey design and administration has been inspired by the "Child-robot Interaction in the Wild" study by Ros et al. [1], where child users' perceptions of robots were assessed. In their study, participants used tactile exploration and expressed interest in realistic aspects of the robots, highlighting the importance of physical engagement.

Our survey questions will be tailored to the cognitive and communication needs of children. They will assess children's satisfaction with the system, their ability to interact with Nao effectively, and their feelings of empowerment in selecting and wearing clothing. We will also inquire about their experiences with the system's recommendations and whether they found them helpful in their daily routines. This approach is rooted in the positive reception of interactive and engaging elements observed in the study by Ros et al. [1]. Here are examples of questions we may ask them:

- “Do you find it challenging to differentiate between different colors of clothing?”
- “Have you ever faced difficulties matching your clothes based on the weather or the occasion?”
- “How confident are you in your clothing choices? Do you worry about making mistakes?”
- “If you could ask a friendly robot for help with choosing your clothes, what would you like it to do for you?”
- “Would it be fun for you to interact with a robot-like Nao when selecting your clothes?”
- “Can you imagine any specific situations where having a robot's assistance in choosing clothes would be helpful to you?”

As our project progresses, we plan to extend these surveys to encompass a broader demographic range, including older individuals with color vision deficiencies. This expansion will help us gauge the applicability of our system across different age groups and gather feedback from a diverse set of participants. Surveys for older individuals will encompass questions regarding the system's ease of use, its influence on daily clothing choices, and its impact on independence and confidence:

- “What strategies have you employed to overcome these challenges in your daily life?”
- “Can you imagine using a robot-like Nao for assistance in clothing selection and coordination?”
- “Are there specific features or functions you believe are essential in a clothing-assistance robot?”
- “Have you ever received assistance in clothing selection before, and if so, how did it impact your daily life?”
- “Would you be open to trying new technologies like a clothing-assistance robot to improve your clothing choices and independence?”

The data collected from these surveys will be analyzed to identify trends, common themes, and areas for improvement. Qualitative responses will be categorized and coded, allowing us to extract meaningful insights. Meanwhile, quantitative data will enable the assessment of user satisfaction, system effectiveness, and overall user experience.

We acknowledge that these insights, based on established methodologies, will be valuable in shaping the ongoing development of our innovative child-robot interaction system, ensuring that it meets the needs of its intended users.

This approach, inspired by the insights from Ros et al.'s research [1], enhances the reliability and relevance of our surveys, aligning them with the unique aspects of our project and its intended users. It allows us to assess the system's impact on the daily lives of colorblind children and, in the future, individuals of different age groups with color vision deficiencies, providing valuable feedback for system refinement and enhancement.

First results

During our initial testing phase, we made substantial progress in the standalone color detection process using a webcam and OpenCV. However, when it came to integrating these capabilities with the Nao robot, we encountered unexpected challenges. Unfortunately, we were unable to execute our color detection algorithm on the Choregraphe simulation or the actual Nao robot due to the deprecated Python 2.7 version used on these platforms.

Regarding the weather API, we achieved successful outcomes when testing it in isolation. However, incorporating this functionality with the Nao proved to be more intricate. We encountered connectivity issues that prevented the robot from accessing the internet to retrieve weather data.

Despite these challenges, we made substantial progress in establishing a dialogue with the Nao. The robot was able to comprehend and respond to user queries and requests effectively.

Dialogue expected during user-Nao interaction:

User: "Hello, Nao. How are you today?"

Nao: "Hello! I'm doing great. How can I assist you today?"

User: "I need help getting dressed. Can you suggest something for me?"

Nao: "Of course! Do you have any ideas in mind?"

User: "Yes."

Nao: "Sure, let me take a look. Could you please show me the clothing you have in mind?"

At this point, the user proceeds to show one clothing item to the Nao robot's camera.

Nao: "I see you have a blue T-shirt. It's a great choice, but considering the current weather, it's

quite chilly outside. How about adding a light jacket to stay warm?"

User: "That sounds like a good idea. What do you suggest for a jacket?"

Nao provides suggestions for suitable jacket options based on the user's clothing choice and the weather conditions. The user and Nao engage in a collaborative decision-making process, where the robot's recommendations are tailored to the individual's preferences and weather-appropriate clothing.

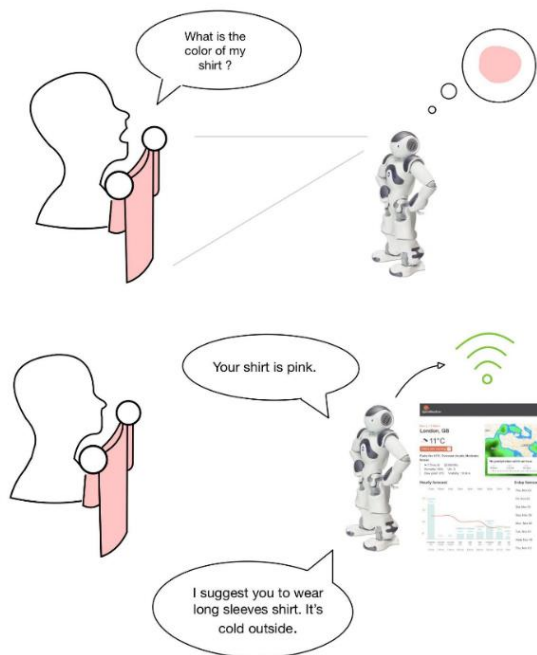


Figure 6: Presentation of how our Nao-based system works

During our testing phase, we observed this interaction between the user and the Nao robot, highlighting the system's ability to effectively engage in a conversation, and understand user requests. The exchange exemplified the system's potential to assist colorblind children in selecting clothing that aligns with their style and the prevailing weather conditions.

However, it's important to note that during the testing phase, we encountered challenges related to ambient noise levels in the testing environments. High levels of background noise in these areas occasionally disrupted the interaction between the user and Nao. Despite these challenges, our initial results demonstrated the feasibility of the system

and motivated us to refine its functionality and robustness for real-world usage.

Conclusion

In our pursuit of leveraging robotics and human-robot interaction to empower colorblind children, our Nao-based system has shown promise as an innovative solution for assisting them in making clothing choices aligned with weather conditions. Throughout this project, we have addressed key elements pivotal to the success of such an assistive technology.

Our project was initiated with the vision of enhancing the confidence and independence of colorblind children by combining the Nao robot's capabilities with advanced algorithms to create a user-friendly and engaging solution that surpasses conventional assistive technologies. The achievements, including clothing color and type detection, clothing recommendations, and facilitating user-robot interaction, are notable.

However, we encountered challenges during our project, primarily related to the integration of technology with the Nao robot, complicated by deprecated software versions and module disparities. These experiences emphasize the need for adaptability and robustness in real-world applications. Moreover, issues such as ambient noise complexities underscore the challenges of creating assistive technology for diverse environments.

It's important to note that we conducted this project under significant time constraints, with only four weeks to achieve initial results. Limited opportunities for real-world testing and extensive simulator-based development were the reality. These constraints highlight our team's dedication and resourcefulness in pushing boundaries within tight timeframes.

Critical Reflection

Our project journey has been marked by remarkable achievements, but it is equally important to reflect on areas where improvement is needed.

The reliance on simulators for most of our development and testing introduced certain limitations. While simulators provide a controlled environment for early-stage development, they do not fully replicate the intricacies and challenges presented in real-world scenarios. Future iterations

of our project could greatly benefit from an increased focus on real-world testing, ensuring that the system's performance is robust and adaptable across diverse settings.

Moreover, we acknowledge the need for more extensive and rigorous evaluation methodologies. While our initial results were promising, a more comprehensive evaluation with colorblind children and a broader user base should be a priority. Gathering feedback and insights from our intended users is crucial to ensuring that the system effectively addresses their unique challenges and aligns with their preferences.

Looking to the future, the refinement and expansion of our system remain on the horizon. The challenges encountered during the integration phase, especially related to software versions and module compatibility, have illuminated the importance of future-proofing our system. This can be achieved by selecting software and platforms that are adaptable to ongoing updates and developments.

Additionally, exploring more advanced machine-learning techniques for color recognition and refining the recommendation algorithms will enhance the system's overall performance. The integration of new technologies and advancements in HRI can further augment the capabilities of our system, making it more user-friendly and versatile.

In conclusion, our critical reflection underscores the importance of addressing challenges related to time constraints, simulator usage, and evaluation methods. While our project has made considerable progress, there is always room for improvement and innovation. We are committed to a future where our system offers comprehensive support for colorblind children, fostering their independence, and enhancing their daily lives.

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